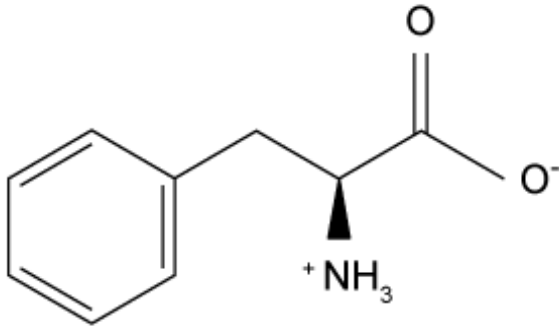


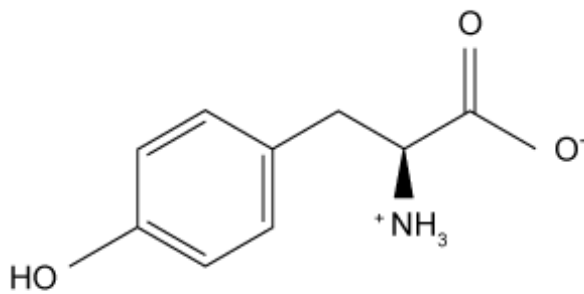
The enzyme phenylalanine hydroxylase catalyzes the conversion of phenylalanine to tyrosine. The disease phenylketonuria (PKU) is caused by an inactivating mutation in the gene encoding phenylalanine hydroxylase. Which structure corresponds to the amino acid that will NOT form as a result of PKU?

☐ A.



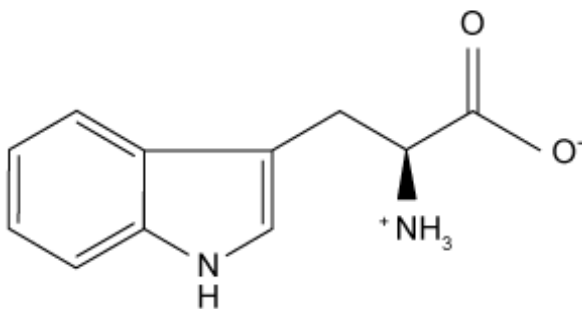
[6%]

☒ B.



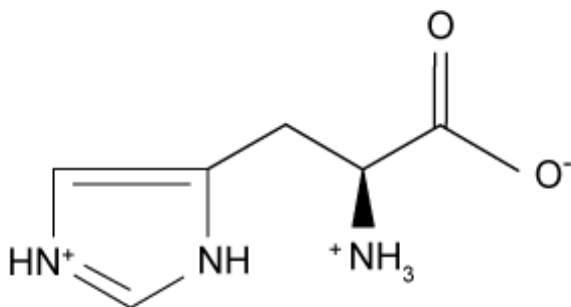
[82%]

☐ C.



[3%]

☐ D.

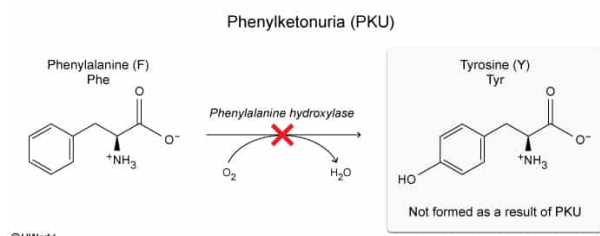


[7%]

Correct

81% answered correctly

Explanation:



Amino acids are important biological molecules that make up proteins and are a source of metabolic energy through oxidative degradation. Amino acids are **made up** of an **amino** group, a **carboxyl** group, and a **side chain** all bonded to the same carbon atom (known as the **α -carbon**). Because amino acid structures differ only by the attached side chain, amino acids can be grouped together by their **side-chain classification**, which relates to the side-chain functional group. Accordingly, amino acid classifications include aromatic, hydrophilic or hydrophobic, acidic or basic, and charged (positive or negative) or uncharged.

Phenylketonuria is characterized by a lack of activity of phenylalanine hydroxylase, which catalyzes the hydroxylation of phenylalanine to tyrosine. Because the enzyme is not functioning properly, tyrosine, which is characterized as an aromatic amino acid with a **phenol functional group**, is not formed.

(Choice A) This aromatic amino acid is **phenylalanine**; it builds up in the blood because it cannot be hydroxylated to form tyrosine as a result of phenylketonuria.

(Choices C and D) **Tryptophan** and **histidine** are also aromatic amino acids, but they are not involved in the hydroxylation catalyzed by phenylalanine hydroxylase; therefore, formation of these amino acids is not affected by phenylketonuria.

Educational objective:

Amino acids are important biological molecules that make up proteins. Amino acids contain an amino group, a carboxyl group, and a side chain all bonded to an α -carbon. The amino acids can be grouped together based on the classification of the side chains and functional group. Classifications include aromatic, hydrophilic or hydrophobic, acidic or basic, and charged (positive or negative) or uncharged.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 401752

System:

Functional Groups and Their Reactions